



RV College of Engineering®
(An autonomous institution affiliated to VTU, Belagavi)
Department of Master of Computer Applications

Date: 06/06/2025

IV SEMESTER – MCA492L Technical Seminar
Screenless Displays: The Future of Visual Interaction

Synopsis

Screenless display technology is transforming visual interaction by eliminating reliance on traditional physical screens. This innovative approach enables digital content to seamlessly integrate with the physical environment whether projected into open space, overlaid onto real-world surfaces, or transmitted directly onto the human retina. Employing advanced methods such as holography, augmented reality (AR), and retinal projection, these systems facilitate immersive, hands-free visual experiences. The shift toward a post-screen paradigm promotes more intuitive, integrated, and unobtrusive means of accessing and interacting with digital information across diverse applications

Recent advancements in screenless display technology, including retinal projection systems, deliver high-resolution visuals directly onto the retina, thereby removing the necessity for physical display panels. Concurrently, holographic displays and AR devices are evolving toward more compact, intelligent designs capable of rendering three-dimensional content and real-time data overlays. The fusion of artificial intelligence, eye-tracking, and spatial computing enables these systems to dynamically tailor visual output according to contextual variables and detected user intent or emotion. Emerging platforms such as Mojo Lens and Microsoft HoloLens exemplify early progress toward fully immersive and responsive screenless ecosystems.

In the near future, screenless display technology is anticipated to transform key sectors such as healthcare, education, defense, and entertainment. Retina-based displays can facilitate real-time data visualization in surgical environments, while holographic systems may enable interactive three-dimensional educational content. Smart wearable devices, including contact lenses with embedded display capabilities, are projected to provide seamless access to digital information without the need for physical interfaces. As the boundaries between the digital and physical environments continue to blur, screenless display technology is positioned to play a central role in redefining the methods of visual communication and information interaction.

Student	Internal Guide	External Evaluator	Director
Rajesh C U (1RV23MC080)	Dr. Deepika K Associate Professor Department of MCA		Dr. Jasmine K S Associate Professor & Director Department of MCA